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(54) LOW-PROFILE ELECTRIC BED

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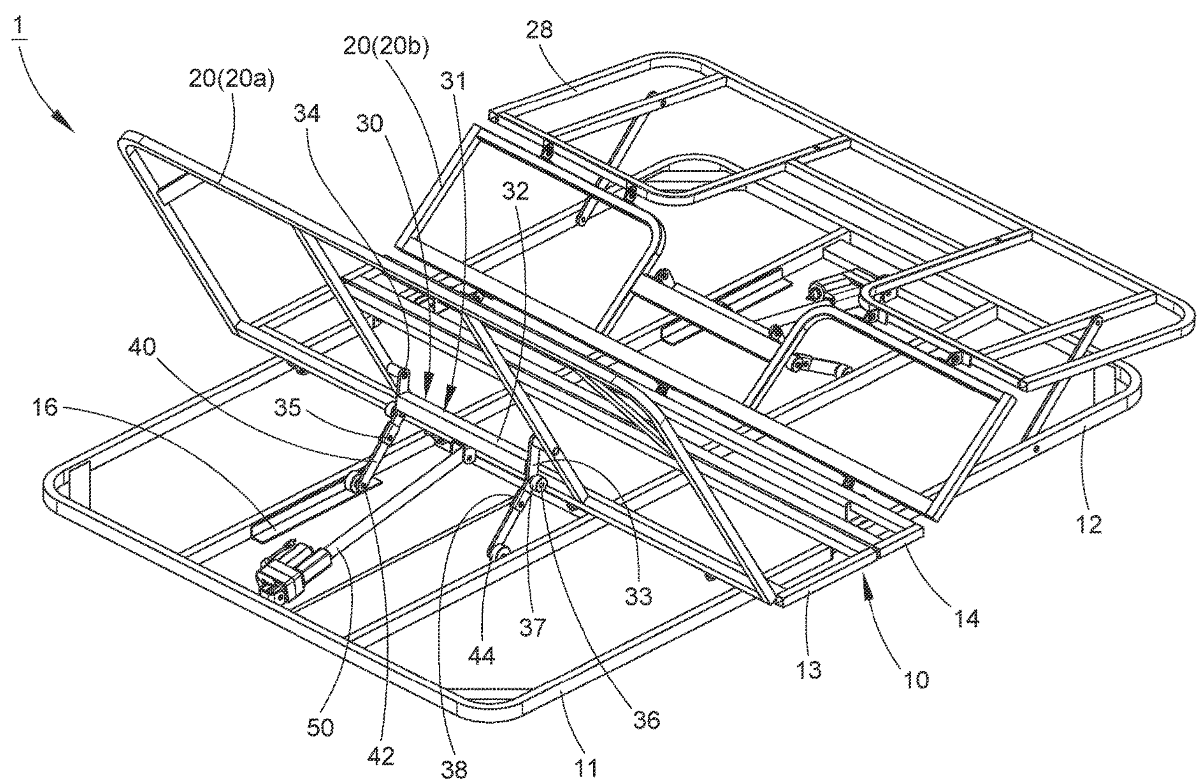
(57) **ABSTRACT**

A low-profile electric bed includes a stationary frame having two slide rails, an upper frame pivotally connected with the stationary frame, a push frame having a swivel rack pivotally connected with the upper frame and two moving arms pivotally connected with swivel rack, and an actuator having two ends pivotally connected with the stationary frame and the upper frame, respectively. The actuator is arranged in a way that when the actuator extends, the upper frame is driven to upward swing relative to the stationary frame. As a result, the low-profile electric bed may have a reduced thickness to minimize packing volume thereof while maintaining a normal function.



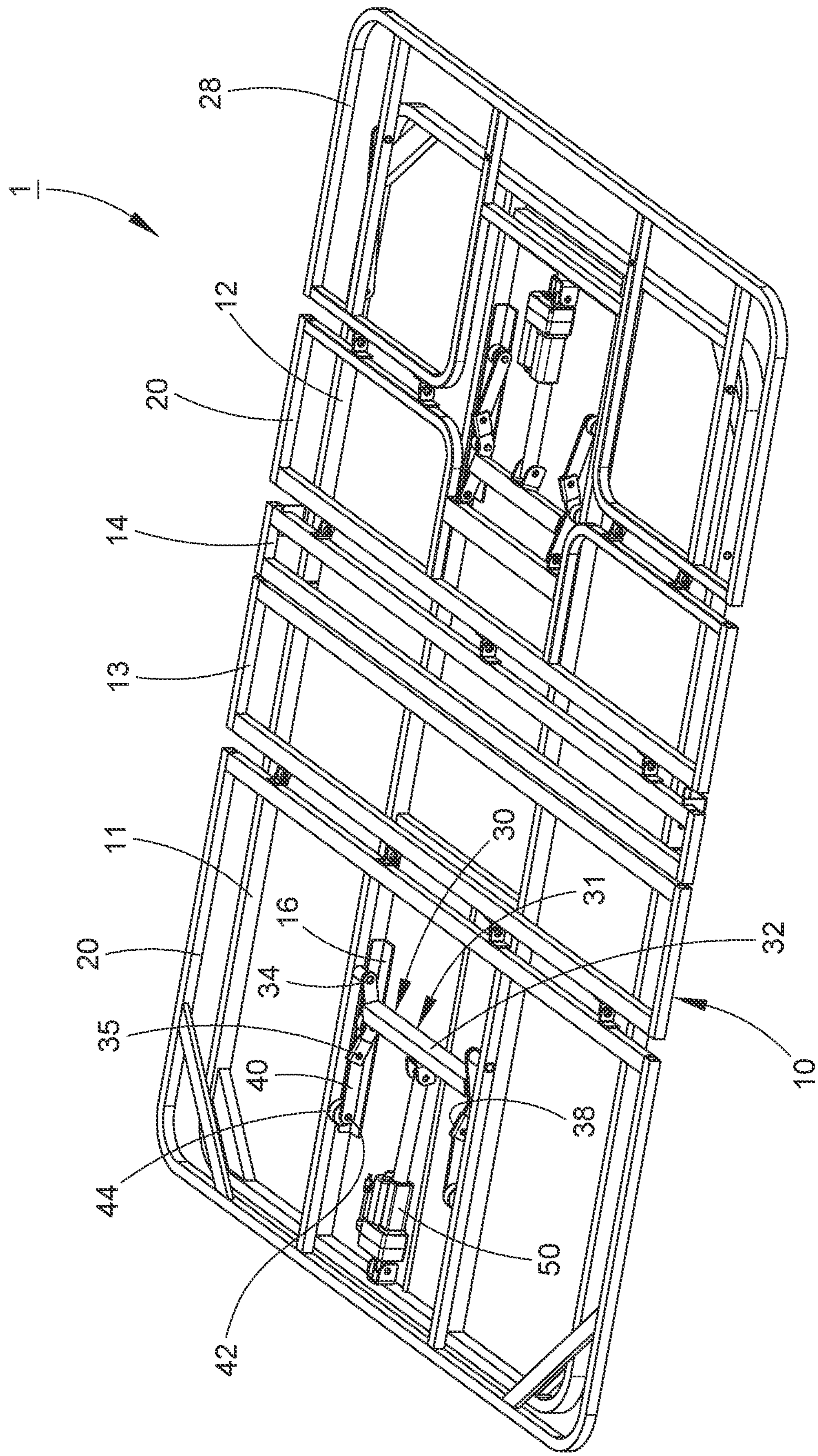


FIG. 1

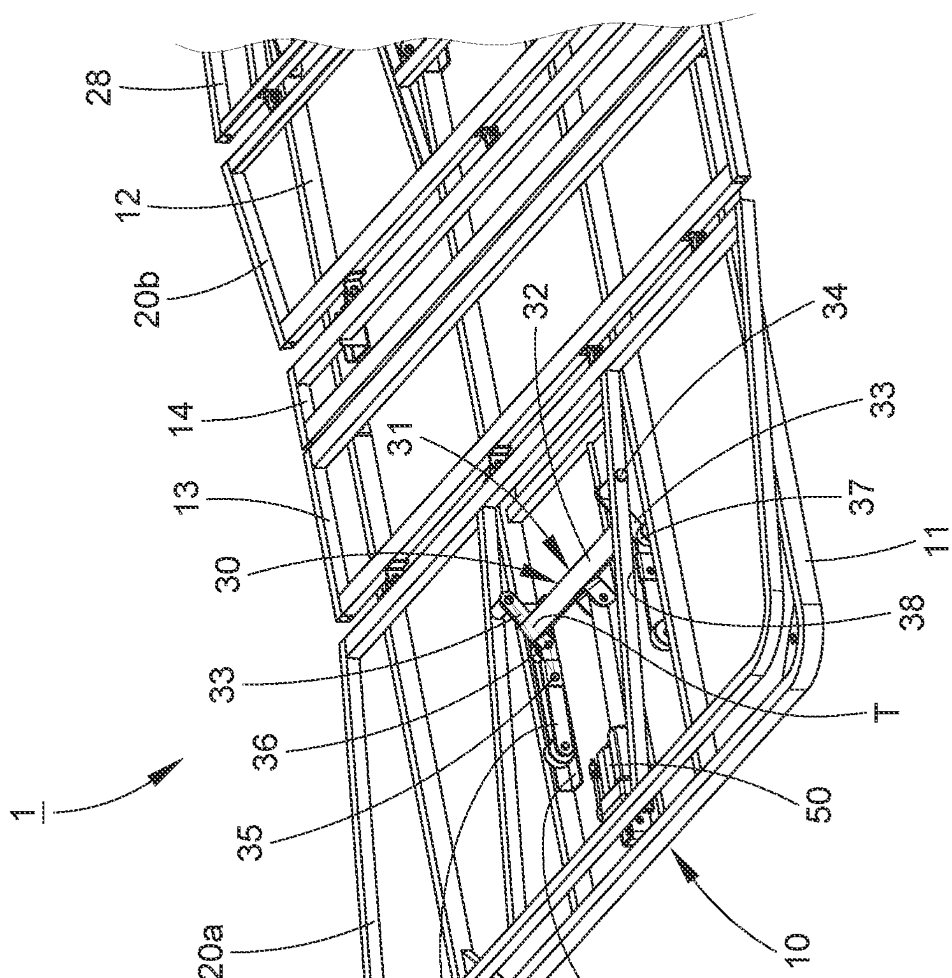
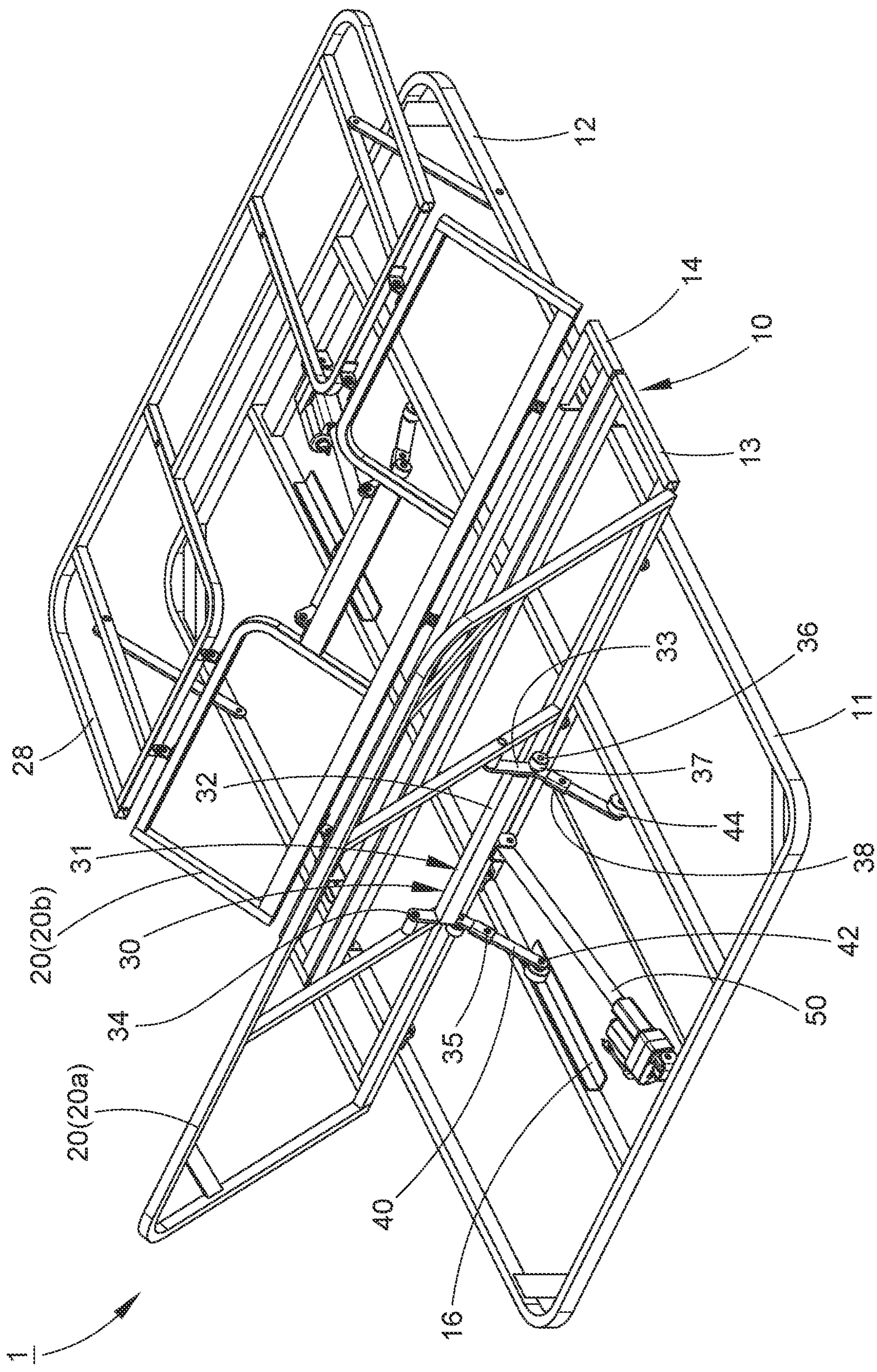


FIG. 2


12
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20(20b)
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20(20a)
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1

FIG. 3

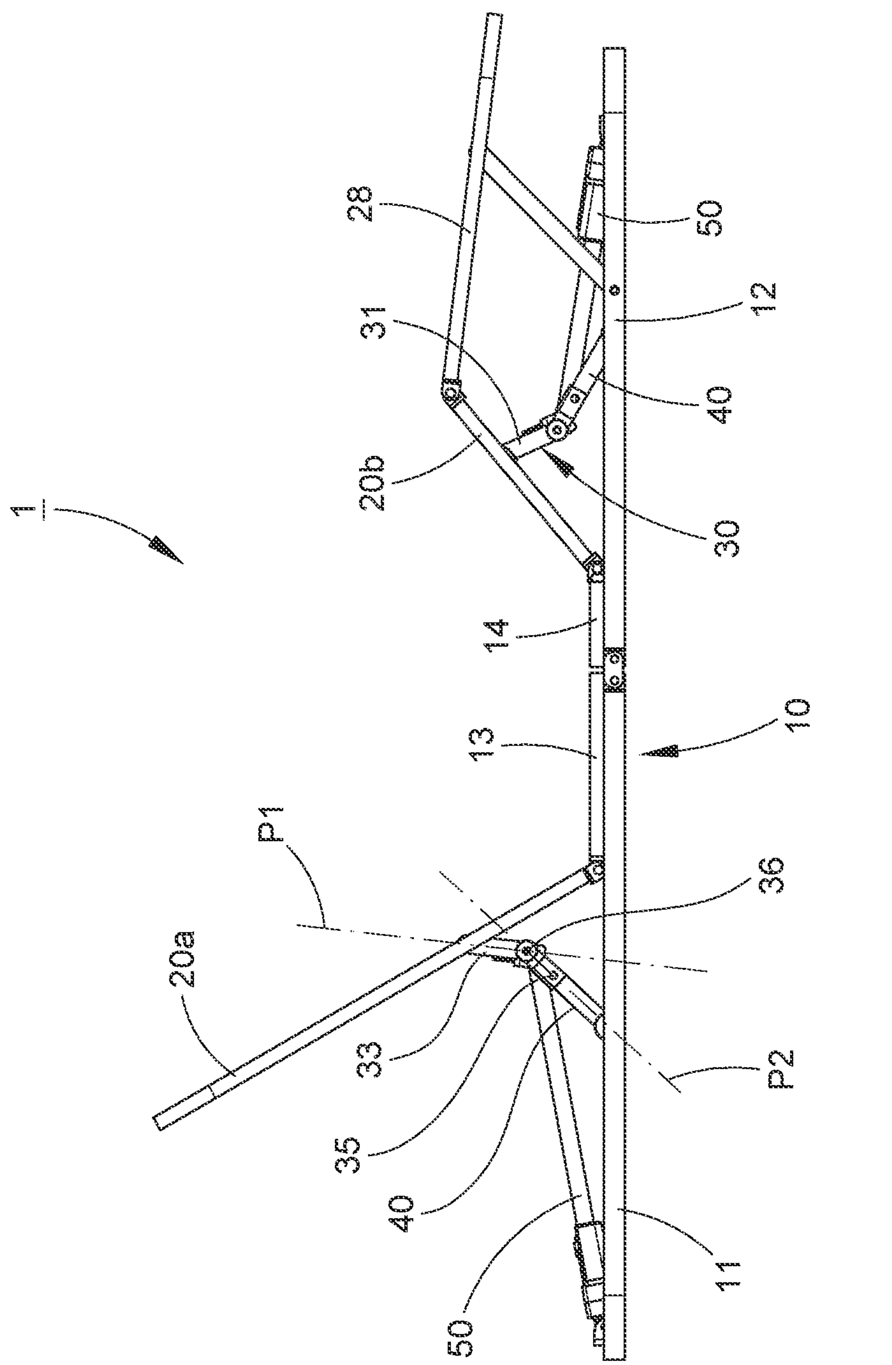

FIG. 4

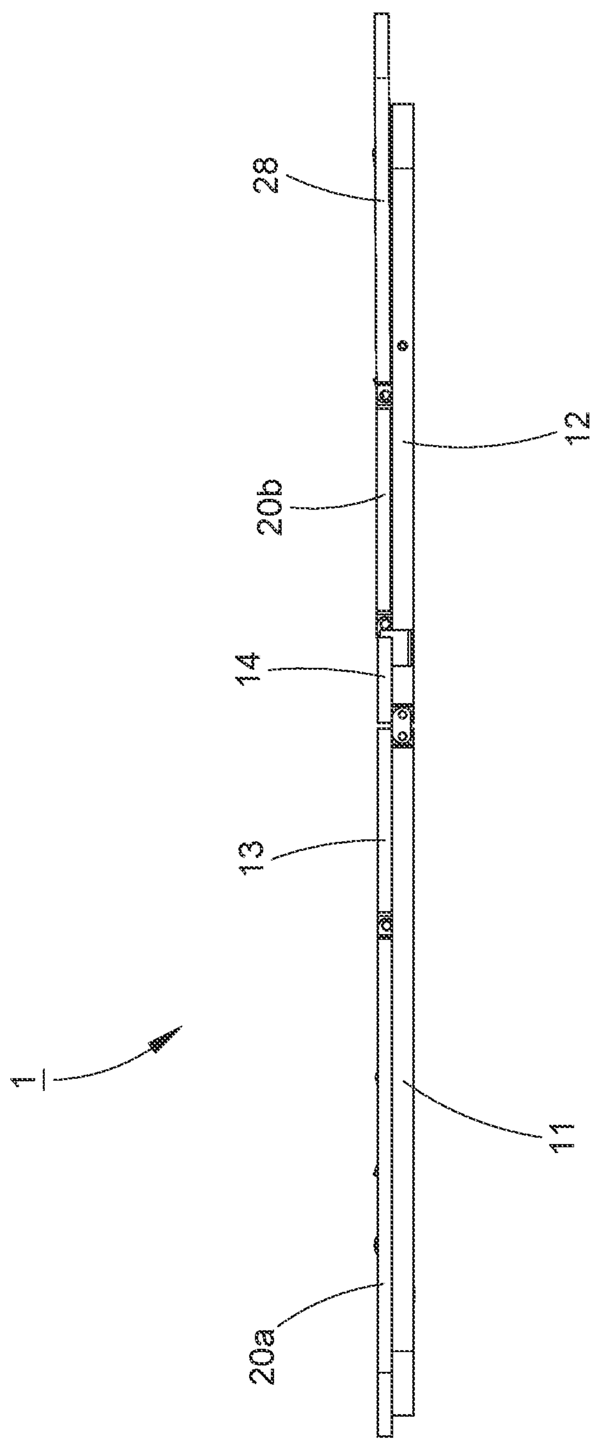

FIG. 5

LOW-PROFILE ELECTRIC BED

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0001] The present invention relates generally to an electric bed and more particularly, to a low-profile electric bed having a thin thickness.

2. Description of the Related Art

[0002] Conventional electric beds are basically composed of essential parts, such as base frame and back, buttock, thigh and lower leg supporting frames located on the base frame, as well as two actuators, such as electric cylinders, mounted to the base frame for driving the back supporting frame and the thigh supporting frame to swing respectively. To get a longer force arm, the back supporting frame and the thigh supporting frame are usually and respectively provided with a downwardly extending support arm to be connected with one of the two actuators, such that the actuators may generate sufficient torque to drive the back supporting frame and the thigh supporting frame to swing.

[0003] Since the two support arms each have a considerable length, they may extend downward beyond the bottom surface of the base frame. If the base frame is intended to be placed on the ground or bed surface, it needs to have sufficient thickness to accommodate these two support arms. As a result, the packing volume of a conventional electric bed is larger, leading to higher transportation and storage costs. Therefore, reducing the packing volume of an electric bed without compromising its normal functionality becomes a technical challenge to be addressed in the industry.

SUMMARY OF THE INVENTION

[0004] The present invention has been accomplished in view of the above-noted circumstances. It is an objective of the present invention to provide a low-profile electric bed, which has a thin thickness, thereby reducing the packing volume thereof.

[0005] To attain the above and other objectives, the present invention provides a low-profile electric bed comprising a stationary frame having two slide rails, an upper frame, a push frame, and an actuator. The upper frame has a side pivotally connected with the stationary frame. The push frame comprises a swivel rack and two moving arms. The swivel rack includes a connecting rod and two support arms disposed with two ends of the connecting rod. Each of the two support arms is pivotally connected with the upper frame via a first pivot, pivotally connected with one of the two moving arms via a second pivot, and pivotally connected with a push wheel via a third pivot in a way that the third pivot is located between the first pivot and the second pivot, and the first, second and third pivots are located at three vertices of a triangle respectively. Each of the two support arms has a stopping portion configured in a way that when each of the two moving arms is rotated relative to associated one of the two support arms at a predetermined angle, each of the two moving arms is stopped at the stopping portion of the associated one of the two support arms to prevent the two moving arms from further rotation. Each of the two moving arms is pivotally connected with a roller via a fourth pivot. The actuator has two ends pivotally and respectively connected with the stationary frame and the swivel rack in a way that when the actuator gradually extends at a first stage, the two push wheels are in contact with the two slide rails respectively, causing the swivel rack to swing about the third pivot and to drive the upper frame to upward swing relative to the stationary frame, and when the actuator further extends at a second stage, the two moving arms are respectively engaged with the two stopping portions, causing that the two rollers are in contact with the two slide rails respectively, and the swivel rack and the two moving arms collectively swing about the fourth pivot in a way that the two push wheels are disengaged from the two slide rails, and the swivel rack and the two moving arms collectively drive the upper frame to upward further swing relative to the stationary frame. As a result, the low-profile electric bed may have a reduced thickness to minimize the packing volume without affecting the normal function.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] The present invention will become more fully understood from the detailed description given herein below and the accompanying drawings which are given by way of illustration only, and thus are not limitative of the present invention, and wherein:

[0007] FIG. **1** is a perspective view of a low-profile electric bed according to an embodiment of the present invention;

[0008] FIG. **2** is a schematic perspective view of the low-profile electric bed of the embodiment of the present invention, showing the electric bed is in use at a first stage;

[0009] FIG. **3** is another schematic perspective view of the low-profile electric bed of the embodiment of the present invention, showing the electric bed is in use at a second stage;

[0010] FIG. **4** is a schematic lateral view of the low-profile electric bed of the embodiment of the present invention, showing the electric bed is in use at the second stage; and

[0011] FIG. **5** is a lateral view of the low-profile electric bed of the embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0012] The structure and technical features of the present invention will be detailedly described hereunder by an embodiment and accompany drawings. As shown in FIGS. **1-4**, a low-profile electric bed **1** provided by an embodiment of the present invention is composed of a stationary frame **10**, two upper frames **20**, a lower leg supporting frame **28**, two push frames **30**, and two actuators **50**.

[0013] The stationary frame **10** is the stationary part of the low-profile electric bed **1** during use, comprising a front base frame **11**, a rear base frame **12** pivotally connected to the front base frame **11**, a front buttock supporting frame **13** fixed to the top side of the front base frame **11**, and a rear buttock supporting frame **14** fixed to the top side of the rear base frame **12**. The front and rear buttock supporting frames **13** and **14** are used to support the mattress (not shown) at the user's buttock position. The stationary frame **10** has four slide rails **16** extending in a front-and-rear direction. In this embodiment, the so-called front-and-rear direction refers to the direction passing through the head and the tail of the bed. Two of these slide rails **16** are disposed on the front base frame **11**, and the other two are located on the rear base frame **12**.

[0014] The two upper frames **20** are respectively and pivotally connected on one side to the stationary frame **10**. In this embodiment, one upper frame **20***a* serves as a back supporting frame, and is pivotally connected at its rear side to the front buttock supporting frame **13** and used to support a mattress for the user's back position. The other upper frame **20***b* serves as a thigh supporting frame, and is pivotally connected at its front side to the rear buttock supporting frame **14** and used to support the mattress for the user's thigh position. The lower leg supporting frame **28** is pivotally connected at its front side to the rear side of the thigh supporting frame. Since the two upper frames **20**, the two push frames **30**, and the two actuators **50** can be divided into two sets with the same structure, for simplification, the following description will use the back supporting frame as an example to illustrate the structural relationships between the upper frame **20** and other components.

[0015] Taking the push frame **30** located on the left side of FIG. **1** for example, the push frame **30** includes a swivel rack **31** and two moving arms **40**. The swivel rack **31** has a connecting rod **32** extending in a left-and-right direction, and two support arms **33** extending in the front-and-rear direction and being connected with two ends of the connecting rod **32**, respectively. Each support arm **33** is pivotally connected to the upper frame **20** via a first pivot **34**, to an associated moving arm **40** via a second pivot **35**, and to a push wheel **37** via a third pivot **36**. The third pivot **36** is located between the first pivot **34** and the second pivot **35**, and the first, second and third pivots **34**, **35** and **36** form three vertices of an imaginary triangle T. Each of the two ends of the connecting rod **32** is positioned between paired one of the first pivots **34** and one of the third pivots **36**. Each support arm **33** has a stopping portion **38** adjacent to the second pivot **35**. When the moving arm **40** rotates relative to the associated support arm **33** to a predetermined angle, the moving arm **40** will be stopped by the stopping portion **38** to prevent the moving arm **40** from further rotation. Each moving arm **40** is pivotally connected to a roller **44** via a fourth pivot **42**.

[0016] As shown in FIG. **4**, the first, second and third pivots **34**, **35** and **36** are configured in a way that there is a first imaginary surface P**1** passing through the first pivot **34** and the third pivot **36**, and a second imaginary surface P**2** passing through the second pivot **35** and the third pivot **36**. Specifically, the first pivot **34** and the third pivot **36** are situated on the first imaginary surface P**1**, and the second pivot **35** and the third pivot **36** are situated on the second imaginary surface P**2** that is defined with the first imaginary surface P**1** an included angle ranging from 130° to 160°. In this embodiment, the aforesaid included angle is 144°; however, through actual testing, the included angle ranging from 130° to 160° may satisfy the requirement to achieve the same effective function. When the moving arm **40** swings upward and is stopped by the stopping portion **38**, the fourth pivot **42** is just located on the second imaginary surface P**2**. However, in another embodiment, the position of the moving arm **40** relative to the swivel rack **31** is not restricted when the moving arm **40** is stopped by the stopping portion **38**.

[0017] Regarding the actuator **50** located on the left side of FIG. **1**, the actuator **50** is, in this embodiment, an electric cylinder, i.e., a rod type linear actuator, having two ends pivotally connected to the front base frame **11** of the stationary frame **10** and the connecting rod **32** of the swivel rack **31**. In another embodiment, the actuator **50** may be realized as a pneumatic cylinder, hydraulic cylinder, or other suitable actuator, and moreover the actuator **50** may be pivotally connected to other parts of the swivel rack **31**.

[0018] In an initial state during actual operation, as shown in FIG. **1**, the upper frame **20** is parallel to the stationary frame **10**. At this state, the distance between the third pivot **36** and the slide rail **16** is less than the distance between the first pivot **34** and the slide rail **16** and less than the distance between the second pivot **35** and the slide rail **16**. In other words, the third pivot **36** is closer to the slide rail **16** than the first pivot **34** and the second pivot **35** are, and the two push wheels **37** are in contact with the two slide rails **16**. When the actuator **50** gradually extends to prolong its length, the two push wheels **37** maintain in contact with the two slide rails **16**, causing the swivel rack **31** to swing about the third pivot **36** and simultaneously to drive the upper frame **20** to upward swing relative to the stationary frame **10**, as shown in FIG. **2**, representing the first stage. When the actuator **50** continuously extends to prolong its length further, the two moving arms **40** will swing relative to the two support arms **33** until the two moving arms **40** are stopped by the two stopping portions **38**. At this point, the fourth pivots **42** are located on the second imaginary surface P**2**, the two rollers **44** are in contact with the two slide rails **16**, and the swivel rack **31** and the two moving arms **40** collectively swing about the fourth pivots **42**, causing that the two push wheels **37** are disengaged from the two slide rails **16**, and the swivel rack **31** and the two moving arms **40** collectively drive the upper frame **20** to upward swing relative to the stationary frame **10**, as shown in FIGS. **3** and **4**, representing the second stage.

[0019] By means of the structural design of moveable joints of the push frame **30**, the height of the push frame **30** can be limited between the bottom side of the stationary frame **10** and the top side of the upper frame **20**. Without the need of using a larger actuator, sufficient torque can be applied to drive the upper frame **20** to swing upward. This design allows the overall thickness of the low-profile electric bed **1** to be only about 5-6 centimeters, as shown in FIG. **5**. Compared to conventional structures of prior arts, the low-profile electric bed **1** provided by the present invention effectively reduces the overall thickness thereof so as to minimize the packing volume thereof, thereby achieving the objectives of the present invention.

[0020] Based on the technical features of the present invention, various modifications to the low-profile electric bed **1** may be made. For example, the shapes of the stationary frame **10** and the upper frame **20** may be changed, and the connecting rod **32** may have a different shape, such as a plate shape, as long as it can connect the two support arms **33**. Such variations are not to be regarded as a departure from the spirit and scope of the invention, and all such modifications as would be obvious to one skilled in the art are intended to be included within the scope of the following claims.

What is claimed is:

1. A low-profile electric bed, comprising:

a stationary frame having two slide rails;

an upper frame having a side pivotally connected with the stationary frame;

a push frame comprising a swivel rack and two moving arms; wherein the swivel rack includes a connecting rod and two support arms disposed with two ends of the connecting rod; wherein each of the two support arms is pivotally connected with the upper frame via a first pivot, pivotally connected with one of the two moving arms via a second pivot, and pivotally connected with a push wheel via a third pivot in a way that the third pivot is located between the first pivot and the second pivot, and the first, second and third pivots are located at three vertices of a triangle respectively; wherein each of the two support arms has a stopping portion configured in a way that when each of the two moving arms is rotated relative to associated one of the two support arms at a predetermined angle, the each of the two moving arms is stopped at the stopping portion of the associated one of the two support arms to prevent the each of the two moving arms from further rotation; wherein each of the two moving arms is pivotally connected with a roller via a fourth pivot; and

an actuator having two ends pivotally and respectively connected with the stationary frame and the swivel rack in a way that when the actuator gradually extends at a first stage, the two push wheels are in contact with the two slide rails respectively, causing the swivel rack to swing about the third pivot and to drive the upper frame to upward swing relative to the stationary frame, and when the actuator further extends at a second stage, the two moving arms are respectively engaged with the two stopping portions, causing that the two rollers are in contact with the two slide rails respectively, and the swivel rack and the two moving arms collectively swing about the fourth pivot in a way that the two push wheels are disengaged from the two slide rails, and the swivel rack and the two moving arms collectively drive the upper frame to upward further swing relative to the stationary frame.

2. The low-profile electric bed as claimed on claim **1**, wherein one of the two ends of the actuator is pivotally connected with the connecting rod of the swivel rack; each of the two ends of the connecting rod is located between one of the two first pivots and one of the two third pivots.

3. The low-profile electric bed as claimed in claim **1**, wherein when the upper frame is parallel to the stationary frame, a distance between the third pivot and the slide rail is less than a distance between the first pivot and the slide rail and less than a distance between the second pivot and the slide rail.

4. The low-profile electric bed as claimed in claim **1**, wherein the first pivot and the third pivot are situated on a first imaginary surface, and the second pivot and the third pivot are situated on a second imaginary surface that is defined with the first imaginary surface an included angle ranging from 130° to 160°.

5. The low-profile electric bed as claimed in claim **4**, wherein when the moving arm is stopped by the stopping portion, the fourth pivot is situated on the second imaginary surface.

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